

Dhairya Jain

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Education

- **Indian Institute of Technology Bombay** — JEE-Advanced : **1886**, JEE-Mains : **1209** (2018 - 2022)
 - *Bachelor of Science* in **Economics** with a *Minors Degree* in **Operations Research** — CGPA: 8.8/10.0
 - Part of **Dean's List** for outstanding academic performance (2019 - 2020)

Work Experience

Microsoft India (R & D) Pvt. Ltd. | Software Engineer II (L-61) (June 2022 - Present)

Working in **Azure Storage - Blob** team to develop the world's leading cloud-based object-storage system

Azure Storage	- Developing the latest scaled-account feature & SKU conversion enabling 6000+ PB-sized accounts - Contributing to high-throughput feature and garbage collection processes for Block-Blob & Append-Blob - Driving Get-Blob API latency improvements by upto 75% to support high performance blob workloads through extensive C++ code optimizations and implementing highly available caching at table server layer. - Implemented state machine to unvirtualize 300+ LRS/ZRS accounts, driving $\approx 15\%$ operational savings - Led the (un-)virtualization feature for scaled GRS accounts, ensuring data integrity using TLA+ specs, and orchestrating components involved in request hot-path across all of data, control & customer planes
Bing Platform	- Engineered ETL pipelines based on <i>Apache Spark</i>, to develop hardware forecasting solutions over big-data - Designed experiments to find the <i>Cost To Serve</i> per Bing Query, and instrument Bing.exe for un-used code
Tech Stack	C++, C#, Virtualization, Scripting (Python, Powershell, KQL), Databases, TLA+

Mirana Toys Pvt. Ltd. | Software Development Engineer (May 2022 - June 2022)

Part of a start-up creating IOT enabled smart toys for kids

Initiatives	- Implemented a social media engine from scratch featuring user login, post creation, and in-game chatting - Created multiple dockerized micro-services and established inter-service communication using message queues
Tech Stack	Python, Django, RabbitMQ, AWS, S3

WorldQuant Research Pvt. Ltd. | Quantitative Researcher & Developer Intern (May 2021 - July 2021)

Worked on 2 projects in the areas of **quantitative development** using Cython, and **quantitative research** using machine learning

Quant Dev	- Designed & implemented a high-efficiency command-line tool & Python package to visualize & analyze alphas - Achieved 100x speed-up by integrating a C++ backend with a Cython-based Python interface
Quant Research	- Researched latest financial & machine learning literature to identify fresh ideas for Portfolio Optimization - Designed Python APIs to create, back-test & compare different ideas of aggregating multiple alpha signals
Tech Stack	C++, Cython, Concurrency, Multi-threading, Linux, Econometrics, Machine Learning, Investment Research

Google Summer of Code'21 | The R project for Statistical Computing (May 2021 - Aug' 2021)

Contributed to **PerformanceAnalytics**, an **R** package used for portfolio risk analysis

Initiatives	- Created functionality for calculating outlier-insensitive risk estimates in the Capital Asset Pricing Model - Published a vignette on CRAN to exhibit new features and improved unit-test coverage by 11%
Impact	Aided 120K+ risk professionals & researchers in making informed investment choices
Tech Stack	R, Quantitative Finance, Portfolio Management, Multi Factor Models, Robust Regression (Huber, MM etc.)

Leadership Roles

Manager of the Web Team | Student Alumni Relations Cell | IIT-Bombay (May 2019 - Apr' 2021)

Promoted to lead the 5-membered web-team in a **3-tier** body working closely with the Dean of Alumni & Corporate Relations, IIT-B

Initiatives	- Anchored SARC UG orientation to introduce 1000+ fresh first-year students to the organization - Administered & maintained 10+ domains of the body using Nginx reverse proxy over Gunicorn web-servers - Developed an interactive website for Alumination 2021 using 3D graphics by using three.js WebGL renderer - Implemented user-tagging functionality in the Institute Yearbook 2021 to enrich experience of 3k+ users
Impact	- Served 1k+ fresh graduates with personalized yearbooks recounting 20k+ memoirs by automating pdf building - Benefited 3k+ students by ideating 9+ events over the year having a combined budget of INR 1.5+ Million - Eliminated server costs by 50% & server hardware requirements by 70% by optimizing various applications